DAINS Project Report

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1 Introduction

Numerical simulation plays an essential role in modeling many physical phenomena, such as weather, climate, aerodynamics and plasma physics.Any physical phenomena can be well described by three foundational sets of universal balance laws, kinematic relations, and material specific constitutive equations. Together they form a complete systems of partial differential equations that describe motion deformation and internal forces. Solving these equations are quite daunting, limited by computational costs of resolving the smallest spatiotemporal features [1]. The constraints of the widespread use of numerical simulation could be solved by machine learning techniques. Testing correlations between various variables and observations, discovering unexpected patterns in nature, and discovering new scientific laws or even better, making predictions without the availability of such laws are all additional benefits of these techniques [2].

2 Physics-based simulations

2.1 Methods

A two-dimensional elastoplastic finite element model was implemented in FEniCS to generate high-fidelity reference data for the active and deep learning stages. The 1.0×1.0 m domain (thickness 1.0 m) was discretized using a 27×27 grid with second-order interpolation. Displacement-controlled loading was applied at the lateral boundaries using a linear profile in the y-direction, progressing up to a maximum displacement u_0.

The material is a rate-independent bilinear elastoplastic solid with kinematic hardening. Total strain decomposes additively into elastic and plastic contributions:

$$\epsilon = \epsilon^e + \epsilon^p \qquad (1)$$

For the elastic contribution, stress is calculated using the linear constitutive relation:

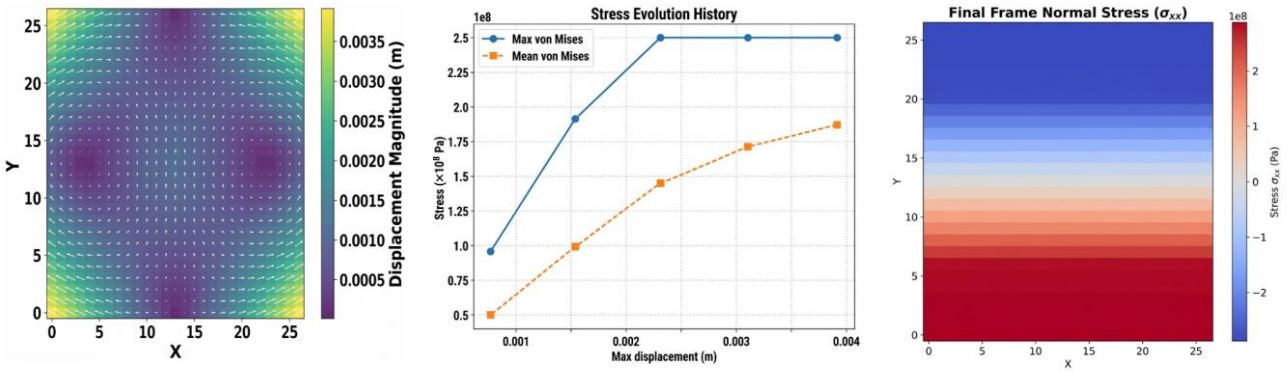
$$\sigma = 2\mu\epsilon^e + \lambda \mathrm{tr}(\epsilon^e)\mathrm{I} \qquad (2)$$

where the Lamé parameters λ and μ depend on Young's modulus E and Poisson's ratio ν. During data generation, E, initial yield stress σ_0, and u_0 were varied. An extremely small tangent modulus ($E_{TAN} = 10^{-9}$ Pa) was utilized to ensure a near elastic-perfectly plastic yield plateau.

Plasticity follows the von Mises yield criterion. A radial return algorithm unconditionally stabilizes the plastic stress correction by projecting the trial elastic stress back onto the yield surface. The non-linear system was solved via an adaptive Newton-Raphson scheme. At each increment, spatial displacement, force, and von Mises stress fields were extracted and mapped onto the grid to form the comprehensive machine learning datasets.

2.2 Results

The kinematic response accurately captures the boundary conditions with maximum corner displacement, demonstrating a clear axis of symmetry at x=0.5 m and an axis of antisymmetry at y=0.5 m. Validating the radial return approach, the peak von Mises stress increases linearly and rigorously caps at the yield barrier (σ_0=250 MPa), while the mean stress continually increases as the plastic zone expands. Spatial analysis reveals a zero-stress neutral axis at y=0.5 m, flanked by an elastic core explicitly bounded within $y \in [0.26, 0.74]$ m. Outside of this core, compressive and tensile plastic zones fully develop at the top and bottom boundaries, respectively.



(a) (b) (c)

Fig. 1. (a) Displacement vectors capturing maximum corner deformation; (b) Stress evolution showing the peak von Mises stress capping at the yield threshold; (c) Final frame normal stress (σ_{xx}) highlighting the neutral axis.

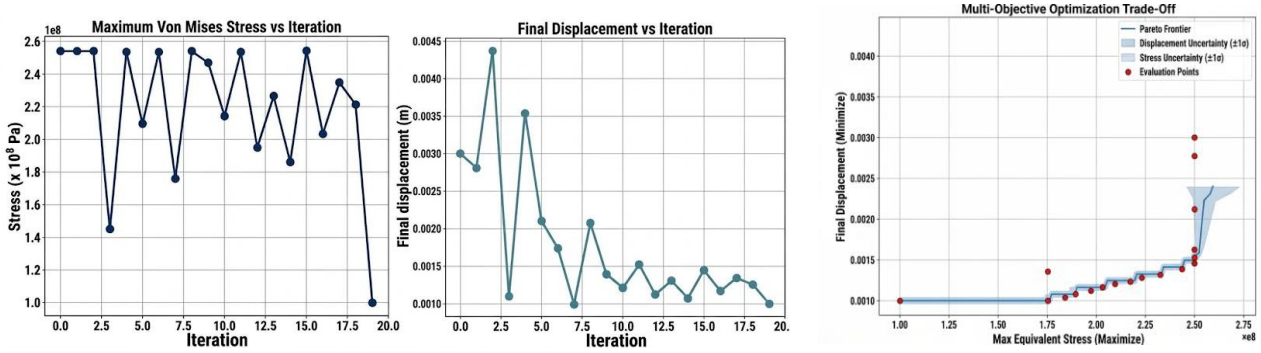
3 Active learning

3.1 Methods

Active learning optimizes data acquisition by proactively selecting the most informative instances to be evaluated when sampling is constrained by high computational costs. The high-fidelity FEniCSx solver served as a deterministic mathematical "oracle" to compute physical responses for sequentially proposed parameters within a Bayesian framework. The multi-objective optimization aimed to maximize peak equivalent von Mises stress while concurrently minimizing maximum final displacement. The continuous design space was bounded by $E \in [50,90]$ GPa and $u_0 \in [1,5]$ mm, keeping the yield stress fixed at $\sigma_0 = 250$ MPa. Implemented via the Ax optimization platform, the active learning routine was executed over 20 iterations. To ensure robust prior formation, the space was initialized at its center, followed by a quasi-random Sobol sequence distribution. Subsequent trials were sequentially proposed using a Gaussian Process (GP) surrogate model to dynamically balance the exploration of high-uncertainty parameter regions with the exploitation of known optimal configurations.

3.2 Results

The sequential evaluations successfully converged to identify a distinct Pareto frontier, highlighting the fundamental trade-off between structural stiffness and deformation. Tracking the metrics across the 20 iterations demonstrates the optimizer's active search, with a notable sharp spike in final displacement required to push the equivalent stress to the 250 MPa yield threshold.



(a) (b)

Fig. 2. (a) Maximum stress and final displacement evaluated across 20 active learning iterations; (b) Multi-objective Pareto frontier showing evaluation points and surrogate uncertainty bounds.

Analysis of the GP surrogate model indicated decoupled parameter sensitivities. The predicted final displacement is directly proportional to the boundary condition u_0, whereas variations in Young's modulus exhibit negligible influence on the final deformation scale. Conversely, the maximum equivalent stress is highly sensitive to the coupled E and u_0 inputs before saturating at the yield limit.

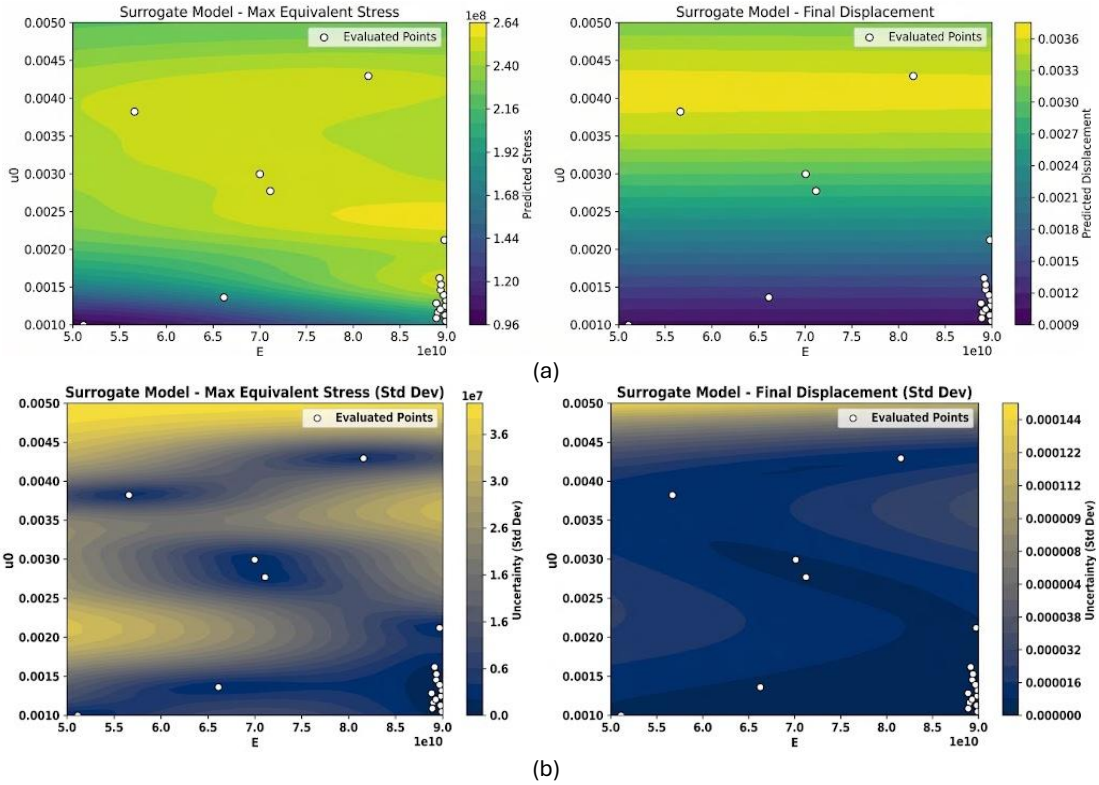


Fig.3.(a) Gaussian Process surrogate mappings for maximum equivalent stress and final displacement; (b) Standard deviation (epistemic uncertainty) of the GP surrogate, highlighting high incertitude in unsampled parameter

4 Deep learning

4.1 Methods

Training generalized deep learning surrogate models required an expanded, independent dataset generated by varying E, σ_0, and u_0. Input features and mechanical outputs were preprocessed using standard z-score normalization to guarantee training stability. The dataset was then split into 70% training, 15% testing, and 15% validation subsets. Standard feedforward topologies are insufficient for inherently path-dependent elastoplasticity [3]. To capture these temporal dependencies, a Long Short-Term Memory (LSTM) recurrent neural network learned the temporal evolution of the system across 6 pseudo-time loading increments to predict scalars like maximum force and peak von Mises stress. The LSTM utilizes a 64-unit layer followed by two Time-Distributed Dense layers (32 and 2 units). The hidden layers utilize ReLU activation functions to efficiently learn non-linear relationships without premature saturation, while the output layer employs a linear activation suitable for continuous regression targets. It was optimized via the Adam algorithm against Mean Squared Error (MSE) loss to strongly penalize large prediction errors.

To predict spatial quantities, a Convolutional Neural Network (CNN) processed FEM outputs as a 27×27 image grid. To capture local and global deformation patterns without downsampling, the architecture applies a fixed 2×3 kernel with exponentially increasing dilation rates ($d \in \{1,2,4,8,16\}$) across five hidden layers. Channel dimensions progress as $32 \rightarrow 64 \rightarrow 32$, culminating in a linear regression layer that preserves the full spatial resolution.

4.2 Results

The machine learning models demonstrated exceptional predictive performance on the unseen testing data. The LSTM successfully captured the dynamic temporal evolution of the mechanical system, achieving a coefficient of determination of $R^2 = 0.998$ for both the maximum reaction force and the maximum equivalent stress scalars. The dilated CNN accurately reproduced the complex spatial von Mises stress contours, achieving an $R^2 = 0.9989$ over the entire computational domain.

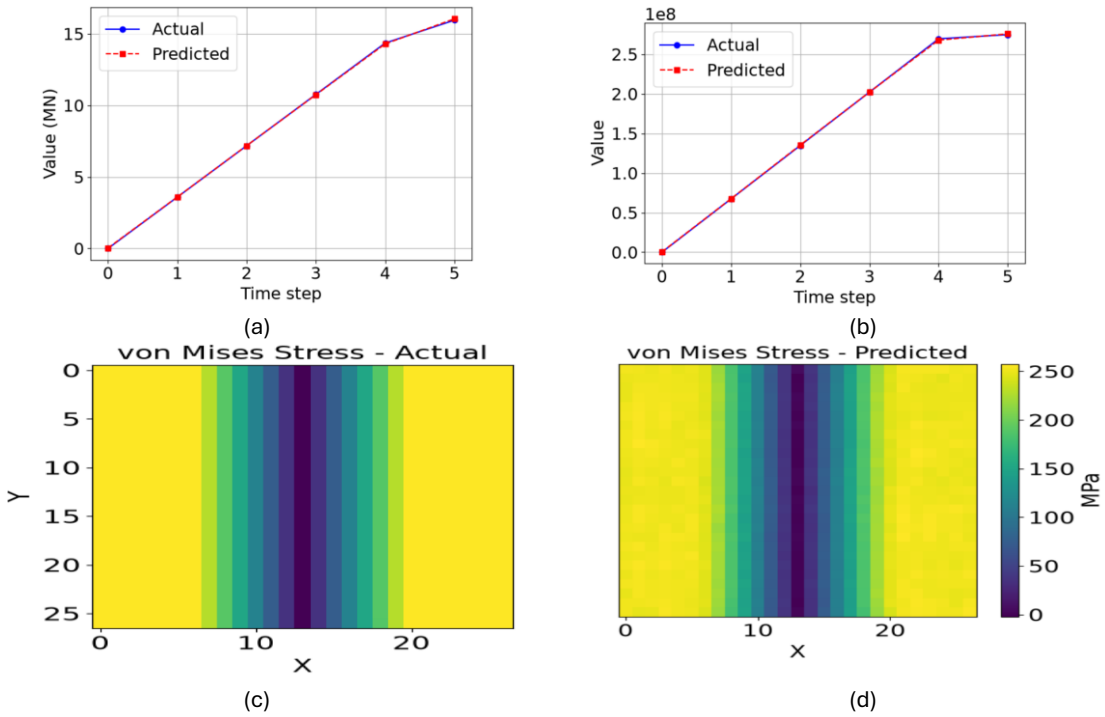


Fig.4. Performance of the deep learning surrogate models evaluated against FEM ground truth data. Top row: LSTM network predictions tracking the temporal evolution of (a) maximum reaction force and (b) maximum von Mises stress across the pseudo-time loading increments, accurately capturing the yield plateau. Bottom row: Spatial comparison of the (c) actual ground-truth von Mises stress field computed via FEniCSx and the (d) predicted stress field generated by the Dilated CNN..

5 Conclusion

This study demonstrates that high-fidelity physics-based simulations can be successfully combined with machine learning frameworks to create highly accelerated and precise surrogate models. These deep learning surrogates overcome traditional processing constraints by addressing the spatial field distributions and path-dependent temporal dynamics of non-linear elastoplastic systems, reducing inference durations from seconds to milliseconds [1][3]. This data-driven system efficiently combines artificial intelligence and traditional solid mechanics, enabling new options for multi-objective optimisation, real-time prediction, and AI-assisted engineering design [3].

References

1. Kochkov, Dmitrii & Smith, Jamie & Alieva, Ayya & Wang, Qing & Brenner, Michael & Hoyer, Stephan. (2021). Machine learning–accelerated computational fluid dynamics. Proceedings of the National Academy of Sciences. 118. e2101784118. 10.1073/pnas.2101784118.
2. Shi, Farui and Li, Minghui and Fantuzzi, Nicholas and Deng, Bozhi and Shang, Delei and Lu, Jun and Xie, Heping, Physics-informed neural network for multiscale mechanical behavior of microstructured composite materials as Cosserat continuum. Available at SSRN: https://ssrn.com/abstract=5430925 or http://dx.doi.org/10.2139/ssrn.5430925.
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